Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/13/2012
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Tufts University		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	04/02/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 4/2/2014, per P. Downey/MSFC (Ed., 2/21/2012) NOTE: Received extension until 4/02/2012 per NSSC (Ed., 4/15/2011) NOTE: Grant may be extended, per P. Downey/MSFC (Ed., 2/18/2011)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Grant Period April 2011-12. • Conducted ESL viscosity tests at ESL-MSFC 7-16 June 2011 • Conducted ESL density tests at ESL-MSFC 9-13 January 2012 • Invited talk at TU-Graz for Prof. Gernot Pottlacher on microgravity testing • Invited talk at International Workshop on Dynamics, Properties and Solidification of Liquid Alloys Symposium in Koln DE 7-9 July 2011 • Delivered presentation at ISPS conference in Bad Godesburg DE 11-15 July 2011 • Attended IWG-4 at ESA HQ in Noordwijk ND 18-19 July 2011 o PARSEC - Peritectic Alloy Rapid Solidification with EM Convection o ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys • Prepared coil deposition summary for FeCo55, FeCo50, FeCr20Ni20 samples • Prepared dust toxicity summary for FeCo55, FeCo50, FeCr20Ni20 samples • Hired Tufts MS student Gabriella String to conduct ground-based EML support testing at Tufts • Hired Post-doctoral researcher Jonghyun Lee to lead the development of convection modeling simulations • Conducted book chapter editorial review for Wiley-VCR as editor of a solidification text involving collaboration with 22 international colleagues from our ESA topical teams 15 June 2011.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Matson DM, Hyers RW, Volkmann T, Fecht H-J. "Phase selection in the mushy-zone: LODESTARS and ELFSTONE projects." Journal of Physics: Conference Series (JPCS) 2011;327:012009. http://dx.doi.org/10.1088/1742-6596/327/1/012009 , Jul-2011
Articles in Peer-reviewed Journals	Hyers RW, Matson DM, Kelton KF, Holland-Moritz D, Volkmann T. "Fluid-flow effects on phase selection and nucleation in undercooled liquid metals." Journal of Physics: Conference Series (JPCS) 2011;327:012013. http://dx.doi.org/10.1088/1742-6596/327/1/012013 , Jul-2011
Dissertations and Theses	Fragoudakis R. "Steel and GFRPs for Beams in Cyclic Loading Application." PhD Thesis, Tufts University, February 2012. , Feb-2012
Dissertations and Theses	Gu R. "Hydrogen Embrittlement Fatigue Properties Using Thermographic Techniques." MS Thesis, Tufts University, May 2011. , May-2011
Papers from Meeting Proceedings	Matson DM. "Controlled Convection during Microgravity Solidification." 50th AIAA Aerospace Sciences Meeting, Nashville, Tennessee, January 9-12, 2012. 50th AIAA Aerospace Sciences Meeting, Nashville, Tennessee, January 9-12, 2012. AIAA paper AIAA-2012-0592. , Jan-2012